FRC 2025 REEFSCAPE — Top Robot Design Deep Dive

A study reference for effective REEFSCAPE design & implementation, drawn from the Spectrum 3847 CAD Collection, team CAD/code releases (Onshape / GrabCAD), and each team's Chief Delphi reveal / CAD-release thread.

The REEFSCAPE design landscape

The game

REEFSCAPE is a coral-and-algae game: robots place CORAL (PVC tubes) on the REEF's four branch levels (L1-L4) and score ALGAE (foam balls) into the PROCESSOR or BARGE NET, then climb a CAGE in the endgame. Most top designs centered on a fast coral cycle (station or ground intake -> elevator/arm -> reef); many top teams ran coral-only and treated algae and deep climb as optional, lower-value choices.

How to use this document

These are the most thoroughly-documented REEFSCAPE robots from the CAD-release set, each with real subsystem detail mined from the team's Chief Delphi reveal/release thread. For every team we capture archetype, the four subsystems (scoring / intake / elevator-lift / drivetrain), vision, public CAD availability, and 'lessons for students'. ★ marks the curated study picks with the richest reproducible documentation. EPA / rank are pending a Statbotics API outage and will be backfilled when it recovers.

Profiled robots at a glance

	Team	Robot	Local EPA	Rec	Archetype	Public CAD/Design
★	**1690 Orbit**	WHISPER	—	—	Closed-belt differential elevator; elite-speed coral cycle	Onshape CAD
★	**1114 Simbotics**	Simbot Suzuki	—	—	Coral-only, station-intake elevator (deliberately minimal scop	Onshape CAD
★	**695 Bison Robotics**	Goldfish	—	—	Fully-autonomous coral scorer; repulsion-field auto-align	Onshape CAD
★	**341 Miss Daisy**	(2025)	—	—	Floor-intake continuous elevator; lantern-gear shoulder + wris	CD thread
★	**1086 Blue Cheese**	Stingray	—	—	Full-capability: 4-level coral, algae net+processor, deep clim	CD thread
★	**1153 Timberwolves**	Lagotto	—	—	1-stage elevator (L1-L3) + algae; ground algae; sub-15 s deep	CD thread
★	**3005 RoboChargers**	Relay	—	—	3-stage elevator + 'laterator' end-effector (coral ejector + a	CD thread
★	**2930 Sonic Squirrels**	(2025)	—	—	Coral robot; deep Choreo auto library + PhotonVision	CD thread
★	**1732 Hilltopper Robotics**	(2025)	—	—	Coral arm/claw; 4x-L4 autonomous; algae handoff to processor	CD thread
★	**2910 Jack in the Bot**	Spectre	—	—	Very compact self-aligning ground coral intake; fast cycle	CD thread
★	**8159 Golden Horn**	Serpentheim	—	—	Coral robot, dual Limelight; 2 regional wins + first Impact Aw	CD thread
★	**190 Gompei and the HERD**	(2025 V2)	—	—	6-motor elevator; star-wheel outtake flips coral up onto L4	CD thread
★	**1156 Under Control**	(2025)	—	—	Swerve coral robot; permanent orientation-PID heading hold; te	CD thread

★ = recommended study pick (downloadable CAD or a full technical binder + a clear design lesson). EPA shown is Local EPA (approx), computed from The Blue Alliance match data.

Team-by-team profiles

★ Study pick **1690 Orbit** WHISPER
ISR
EPA pending

Closed-belt differential elevator; elite-speed coral cycle

SCORING (CORAL / ALGAE) Coral end-effector fed off the differential elevator for L1-L4 placement; among the fastest cycles in the world that season.	**INTAKE** Coral handling tuned for a rapid, repeatable cycle (see CAD).
ELEVATOR / LIFT Closed-belt DIFFERENTIAL elevator — two motors drive a closed belt loop so the carriage height and a second axis are co-controlled; credited to a community design Orbit adapted.	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check CAD)	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A closed-belt differential elevator couples two motions through one belt loop — compact and stiff.
- World-class results come from cycle-time obsession, not feature count.

Onshape CAD: https://cad.onshape.com/documents/76609fe05a6594c5f9c4062a/w/437a97728f1629348e9dd7cf/e/465d3a35c190dab8355f1e5c · CD reveal — WHISPER · CD CAD release

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★ Study pick **1114 Simbotics** Simbot Suzuki
CAN
EPA pending

Coral-only, station-intake elevator (deliberately minimal scope)

SCORING (CORAL / ALGAE) Coral end-effector placing on the reef; team explicitly chose NOT to do ground algae or floor coral.	**INTAKE** Human-player STATION intake for coral (no floor intake by design).
ELEVATOR / LIFT Elevator superstructure; manipulator position recovery was a known failure mode they discuss in the release.	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check CAD)	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Scope discipline: dropping deep climb, ground algae and floor coral let them perfect one cycle.
- Document your failure modes (lost manipulator position) — it teaches more than a highlight reel.

Onshape CAD: https://cad.onshape.com/documents/8782ccd3db870fc41090b5e6/w/88d1084497e586d2c58a4051/e/f1dc3344d36d78907a60d4d9 · CD CAD release

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★ Study pick **695 Bison Robotics** Goldfish **EPA pending**

Fully-autonomous coral scorer; repulsion-field auto-align

SCORING (CORAL / ALGAE) Coral end-effector with fully autonomous scoring across reef levels.	**INTAKE** Autonomous feeder-station pickup (no driver input needed for the cycle).
ELEVATOR / LIFT Elevator superstructure; all mechanism control loops run on-board the motor controllers.	**DRIVETRAIN** Full Kraken-X60 swerve drivebase.
VISION Dual Limelight 4 running MegaTag2; 200 Hz odometry fused with a NavX2 for robust localization.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- MegaTag2 + high-rate odometry (200 Hz) makes full-auto scoring practical.
- A repulsion (potential) field is a clean way to auto-align while avoiding field collisions.
- Running control loops on the motor controllers offloads the roboRIO.

Onshape CAD: https://cad.onshape.com/documents/952ba22fa909a3d0882388da/w/94a37671574438408b83a5fc/e/97e98c3970941d33c368b387 · CD CAD & code release

★ Study pick | **341 Miss Daisy** (2025) | USA-PA | **EPA pending**

Floor-intake continuous elevator; lantern-gear shoulder + wrist; deep climb

SCORING (CORAL / ALGAE) Coral end-effector on a lantern-gear-driven SHOULDER with a 180-degree WRIST for multi-level reef placement.	**INTAKE** Coral FLOOR intake.
ELEVATOR / LIFT Continuous (single-stage continuous) elevator; a software 'superstructure' supervises arm/elevator states to prevent self-collision.	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check CAD/code)	**PUBLIC CAD / DESIGN** CD CAD/Code/Docs release

LESSONS FOR STUDENTS

- A state-supervising 'superstructure' layer prevents the arm from colliding with the robot — model your mechanism states explicitly.
- Floor intake + continuous elevator + 180-degree wrist is a flexible multi-level coral solution.

CD thread: https://www.chiefdelphi.com/t/500739 · CD CAD, code & documentation release · Onshape CAD

★ Study pick | **1086 Blue Cheese** Stingray | USA-VA | **EPA pending**

Full-capability: 4-level coral, algae net+processor, deep climb

SCORING (CORAL / ALGAE) Coral on all four reef levels; algae removed from the reef and scored in both the net and the processor.	**INTAKE** Coral from the human station; algae off the reef.
ELEVATOR / LIFT Elevator superstructure carrying the coral/algae effector.	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check CAD/code)	**PUBLIC CAD / DESIGN** CD CAD & code release

LESSONS FOR STUDENTS

- Study this as an ambition-vs-execution case, not a design exemplar: a genuine do-everything robot (4-level coral + algae net/processor + deep climb) is buildable, but here the scope outran on-field results (EPA rank ~657/3690).
- The lesson is scope discipline — weigh how much capability a team can realistically build, debug, and drive reliably before committing to do-everything.
- Releasing CAD + code together still makes the whole ambitious system reproducible to learn from.

CD thread: https://www.chiefdelphi.com/t/506376 · CD CAD & code release · Onshape CAD

★ Study pick | **1153 Timberwolves** Lagotto | **EPA pending**

1-stage elevator (L1-L3) + algae; ground algae; sub-15 s deep climb

SCORING (CORAL / ALGAE) Coral on L1-L3; algae scored into the processor and barge.	**INTAKE** Ground ALGAE intake (plus coral handling).
ELEVATOR / LIFT Single-stage elevator — simpler than multi-stage rivals, traded top-level reach for reliability.	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check CAD)	**PUBLIC CAD / DESIGN** CD CAD release

LESSONS FOR STUDENTS

- A 1-stage elevator can win districts if the rest of the cycle is fast and reliable.
- A sub-15-second deep climb is a strong endgame insurance policy.

CD thread: https://www.chiefdelphi.com/t/502263 · CD CAD release · Onshape CAD

★ Study pick

3005 RoboChargers Relay

USA-TX

EPA pending

3-stage elevator + 'laterator' end-effector (coral ejector + algae gripper)

SCORING (CORAL / ALGAE) A 'laterator' carries both a coral ejector and an algae gripper; coral is routed THROUGH the elevator to the effector.	**INTAKE** Funnel-assisted coral handling; algae gripper for reef/processor.
ELEVATOR / LIFT 3-stage elevator carrying the laterator.	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check CAD)	**PUBLIC CAD / DESIGN** CD reveal — Relay

LESSONS FOR STUDENTS

- Routing the game piece through the elevator saves a separate hand-off path.
- A 'laterator' (lateral translator) end-effector neatly combines two scoring jobs.

CD thread: https://www.chiefdelphi.com/t/493529 · CD reveal — Relay · Onshape CAD

★ Study pick

2930 Sonic Squirrels (2025)

USA-WA

EPA pending

Coral robot; deep Choreo auto library + PhotonVision

SCORING (CORAL / ALGAE) Coral end-effector for reef placement.	**INTAKE** Coral intake (see CAD).
ELEVATOR / LIFT Elevator superstructure.	**DRIVETRAIN** Swerve.
VISION PhotonVision on two Orange Pi 5 coprocessors; 100+ short Choreo paths stitched into driver-selectable autos.	**PUBLIC CAD / DESIGN** CD CAD & code release

LESSONS FOR STUDENTS

- Build autos from many SHORT Choreo paths so the drive team can compose routines on the fly.
- Dual coprocessors give you redundant / wider-FOV PhotonVision coverage.

CD thread: https://www.chiefdelphi.com/t/505039 · CD CAD & code release · Onshape CAD

★ Study pick

1732 Hilltopper Robotics (2025)

USA-KY

EPA pending

Coral arm/claw; 4x-L4 autonomous; algae handoff to processor

SCORING (CORAL / ALGAE) Coral claw placing on the reef (up to four L4 in auto); algae handed down into the processor.	**INTAKE** Coral intake feeding the claw; algae path with added guarding to keep coral out.
ELEVATOR / LIFT Arm/lift superstructure.	**DRIVETRAIN** Swerve.
VISION QuestNav (Meta Quest headset) used for localization in the 4x-L4 auto.	**PUBLIC CAD / DESIGN** CD reveal (4x L4 QuestNav auto)

LESSONS FOR STUDENTS

- QuestNav (a VR headset's inside-out tracking) is a low-cost, high-rate localization source for autos.
- Design the algae path so it can't accidentally ingest coral.

CD thread: https://www.chiefdelphi.com/t/492785 · CD reveal — 4x L4 QuestNav auto · Onshape CAD

★ Study pick	**2910 Jack in the Bot** Spectre USA-WA	**EPA pending**

Very compact self-aligning ground coral intake; fast cycle

SCORING (CORAL / ALGAE) Coral end-effector for reef placement; high throughput (paired with 9442 at events).	**INTAKE** One of the smallest self-aligning GROUND coral intakes in the game — centers the coral as it ingests.
ELEVATOR / LIFT Elevator superstructure.	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check CAD)	**PUBLIC CAD / DESIGN** CD CAD release

LESSONS FOR STUDENTS

- Passive geometry can self-center a game piece, shrinking the intake envelope.
- A small, reliable ground intake enables a faster, more flexible cycle.

CD thread: https://www.chiefdelphi.com/t/500310 · CD reveal — Spectre · CD CAD release · Onshape CAD

★ Study pick	**8159 Golden Horn** Serpentheim TUR	**EPA pending**

Coral robot, dual Limelight; 2 regional wins + first Impact Award

SCORING (CORAL / ALGAE) Coral end-effector for reef placement.	**INTAKE** Coral intake (see CAD).
ELEVATOR / LIFT Elevator superstructure.	**DRIVETRAIN** Swerve.
VISION Two Limelight cameras (discussed in the release Q&A).	**PUBLIC CAD / DESIGN** CD CAD, code & strategy release

LESSONS FOR STUDENTS

- Pairing CAD + code + STRATEGY documents is rare and valuable — study how strategy drove their design.
- Strong engineering and outreach can come together (2 regionals + first Impact Award).

CD thread: https://www.chiefdelphi.com/t/509509 · CD CAD, code & strategy release · Onshape CAD

★ Study pick	**190 Gompei and the HERD** (2025 V2) USA-MA	**EPA pending**

6-motor elevator; star-wheel outtake flips coral up onto L4

SCORING (CORAL / ALGAE) Outtake with STAR WHEELS that flip the coral upward as it leaves, seating it on L4; a ramp straightens/holds the coral instead of a powered belt.	**INTAKE** Coral intake feeding the ramp.
ELEVATOR / LIFT 6-motor elevator (high power for fast, repeatable height changes).	**DRIVETRAIN** Swerve.
VISION Unconfirmed (check CAD)	**PUBLIC CAD / DESIGN** CD CAD release

LESSONS FOR STUDENTS

- Star wheels can re-orient a game piece during ejection (flip coral up for L4).
- Sometimes a passive ramp beats a powered belt for straightening/holding a piece.

CD thread: https://www.chiefdelphi.com/t/503355 · CD CAD release · Onshape CAD

★ Study pick

1156 Under Control (2025)

BRA

EPA pending

Swerve coral robot; permanent orientation-PID heading hold; tech binder

SCORING (CORAL / ALGAE) Coral end-effector for reef placement.	**INTAKE** Coral intake (see CAD/binder).
ELEVATOR / LIFT Elevator superstructure.	**DRIVETRAIN** Swerve that always runs with an orientation (heading) PID engaged.
VISION Unconfirmed (check binder)	**PUBLIC CAD / DESIGN** CD CAD, code & tech-binder release

LESSONS FOR STUDENTS

- Always-on heading PID makes driving and auto-aim more deterministic.
- A released technical binder is the gold standard for reproducibility.

CD thread: https://www.chiefdelphi.com/t/505398 · CD CAD, code & tech binder · Onshape CAD

Additional well-documented robots for study

Each has a strong public release (CAD, code, binder, or detailed reveal) and a clear design lesson. ★ marks the recommended study picks.

Sources: the Spectrum 3847 CAD Collection and team CAD releases (Onshape, GrabCAD), Chief Delphi reveal / CAD-release threads, and match data from The Blue Alliance. EPA shown is Local EPA (approx), independently computed — not official Statbotics. "Unconfirmed" means a detail was not publicly published, not that it is absent.